

AIMS System - Detailed Class Design Documentation

1. Design Class: Product

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	productID	int	null	Unique identifier for the product
2	title	String	null	Name of the product
3	category	String	null	Product category (Book, CD, DVD, or LP record)
4	value	float	0.0	Base price of the product before VAT
5	price	float	0.0	Selling price of the product
6	quantity	int	0	Available quantity in stock
7	description	String	null	Product description
8	imageUrl	String	null	The URL for the image of the product item
9	barCode	String	null	Unique barcode for the product
10	warehouseEntryDate	Date	null	Date when the product entered the warehouse
11	dimensions	String	null	Physical dimensions of the product (e.g., 10x20x5 cm)
12	weight	float	null	Weight of the product in kilograms
13	warehouseProvince	String	null	province of warehouse storing the product
14	warehouseDistrict	String	null	district of warehouse storing the product
15	warehouseAddress	String	null	detailed address of warehouse storing the product

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	checkProductAvailability	boolean	Checks if the product has sufficient quantity available
2	updateProduct	void	Updates product information in the system
3	checkEligibilityOfRushDelivery	boolean	Checks if product is eligible for rush delivery
4	createProduct	void	Create a new product in the system
5	searchProduct	void	Search for a specific product attributes

Parameter:

- checkProductAvailability(quantity: int): Boolean
 - **quantity:** An integer representing the number of products to check if the available inventory is sufficient.
 - **Return:** true if the available inventory is sufficient, otherwise false.
- updateProduct(productID: int): void
 - **productID:** productID must exist
- searchProduct(searchCriteria: Map<String, Object>): void
 - **searchCriteria:** A map containing search criteria (field names and values)

Exception:

- updateProduct(productID: int): void
 - ProductNotFoundException: If no product with the specified productID is found.
 - ExceedUpdateTimesException: If update more than 30 products a day
 - ExceedUpdatePriceTimesPerDayException: If update price more than twice a day

Method:

How to use parameters/attributes:

- checkProductAvailability(quantity: int): Boolean
 - **Purpose:** This method checks whether the available inventory is sufficient for the requested quantity of a product.
 - **Usage:** If the available stock is at least equal to the requested quantity, the method returns true; otherwise, it returns false.
- updateProduct(productID: int): void
 - **Purpose:** This method updates the details of an existing product based on the provided information.
 - **Usage:** If the product exists, the method updates attributes such as name, price, quantity, etc
- createProduct(): void
 - **Purpose:** This method creates a new product with the provided information.
 - **Usage:** If all required attributes are provided and valid, the product is successfully created and added to the inventory.
- checkEligibilityOfRushDelivery(address: String): Boolean
 - **Purpose:** to check rush delivery support
 - **Usage:** compare 'address' parameter (from delivery information) and address of product (based on information provided by Product Manager)
- searchProduct(searchCriteria: Map<String, Object>): void
 - **Purpose:** Searches for products matching specific criteria.
 - **Usage:** Used to find products based on various attributes.
 - **Processing:**
 1. Constructs a database query based on the search criteria
 2. Executes the query
 3. Returns matching products

State:

- If stock > requested quantity → Available.
- If stock < requested quantity → Not Available.
- If product_id is invalid → ProductNotFoundException is raised.